ثانوية المدينة التعليمية
Education City High School

2.15

Semi-log Plots

Core-to-Challenge Worksheet and AP Practice

AP Precalculus - Unit 2 | ECHS Mathematics | Teacher: Mohammad Abu-Ghuwaleh

Student Information

Name: ______________________ Class: ____________ Date: ____________

Directions

Show mathematical evidence. Retain domain restrictions, label graphs, include units, and write complete interpretations. Calculator use is permitted only when the item explicitly requires approximation, regression, or numerical solving.

Readiness

1. Write one precise sentence explaining this principle: If $y = ab^x$, then $\log y = \log a + x \log b$.

2. Write one precise sentence explaining this principle: On a log-output plot, slope is $\log b$ and intercept is $\log a$.

3. Write one precise sentence explaining this principle: With base 10, $a = 10^{\text{intercept}}$, $b = 10^{\text{slope}}$.

4. Write one precise sentence explaining this principle: A vertical shift $y = k + ab^x$ is not linearized by plotting $\log y$; plot $\log(y - k)$ instead.

Core

5. A semi-log line is $Y = 0.301 + 0.176x$, where $Y = \log_{10} y$. Find the model.

6. What does curvature on a semi-log plot indicate?

7. Why does changing the log base not change whether the transformed data are linear?

8. State what the formula $y = ab^x \Rightarrow \log_{10} y = \log_{10} a + x \log_{10} b$ tells you and name any required restriction.

9. State what the formula $a = 10^c,\ b = 10^m$ tells you and name any required restriction.

10. Create a second representation (table, graph, formula, or context) for one central relationship in Semi-log Plots. State its domain and justify one key feature.

Medium

11. A base-10 semi-log line is $Y = 0.301 + 0.176x$, where $Y = \log_{10} y$. Find the exponential model.

12. A base-10 semi-log line is $Y = 1 + 0.301x$, where $Y = \log_{10} y$. Find the exponential model.

13. A base-10 semi-log line is $Y = 0 + -0.097x$, where $Y = \log_{10} y$. Find the exponential model.

14. A base-10 semi-log line is $Y = 2 + 0.4771x$, where $Y = \log_{10} y$. Find the exponential model.

15. A base-10 semi-log line is $Y = -1 + 0.2x$, where $Y = \log_{10} y$. Find the exponential model.

Hard

16. An exponential has a=5,b=1.2. Find semi-log slope/intercept in base 10.

17. Why do y-values 2,20,200 appear equally spaced on a log-10 axis?

18. A semi-log plot bends upward. What might this suggest?

19. If log base changes from 10 to e, how do line slope/intercept change?

Difficult

20. Data follow y=4+3(2^x). Which transformation yields a line?

21. A semi-log line through (2,1) and (7,2.5) uses Y=log10 y. Find model.

22. Explain why a straight line on a log-log plot indicates a power model rather than an exponential model.

Challenge / HOT

23. Prove any exact exponential dataset with equal x spacing lies on a line in a semi-log output plot.

24. Can negative y-data be shown on a standard logarithmic output axis? Explain alternatives.

AP-Style Multiple Choice

25. For y=ab^x, semi-log slope is

- **(A)** a
- **(B)** b
- **(C)** log b
- **(D)** log a

26. Semi-log intercept is

- **(A)** log a

(B) a
(C) b
(D) 1/a

27. A straight semi-log plot suggests
(A) linear original
(B) exponential original
(C) quadratic original
(D) periodic original

28. Equal vertical log-axis spacing represents equal
(A) differences
(B) ratios
(C) second differences
(D) angles

29. If model includes vertical shift k, linearize
(A) log y
(B) log$(y - k)$
(C) $y - k$ only
(D) $1/y$

30. Log base choice
(A) changes linearity
(B) does not change linearity
(C) makes y negative
(D) changes x-domain

AP-Style Free Response

31. A base-10 semi-log plot of data has line $Y = 0.7 + 0.12x$. (a) Construct original model. (b) Find initial value and one-unit factor. (c) Predict y at x=10. (d) Interpret slope.

32. Data follow $y = 50 + 8(1.4)^x$. (a) Explain why plotting log y vs x is not exactly linear. (b) Give a linearizing transformation. (c) State resulting line.

Reflection

Identify one item where a domain restriction, unit, assumption, or representation changed your reasoning. Explain in 2-3 sentences.

Original ECHS questions. Selected concepts are informed by Pearson and Holt Precalculus; no textbook exercises are reproduced.